SETHURAMAN KARUNANITHI

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EXPERIENCE

One Community Global — Remote
Industrial Engineer (Volunteer) — *Sep 2026 – Present*

- Rebuilding the outdated Highest Good Network user manual across **seven user roles**, starting with volunteers.
- Writing **SOP-style procedures** structured so both a new user and an AI chatbot can follow them step by step.
- Testing every page of the application and fixing **process documentation** where steps did not match the software.

Samsung India Electronics Pvt. Ltd. — Chennai, India
Associate Engineer — Process Engineering — *Oct 2022 – Jun 2024*

- Improved manufacturing processes for safety, quality, productivity and cost — line efficiency **70% to 87%**.
- Cut scrap from **6.3% to 0.78%** and kept **OEE above 85%**, saving **$57K** against a **$112K** yearly cost target.
- Rebalanced takt **14.8s to 4.5s** through time studies, then redesigned floor layout and material flow.
- Found **root cause** on field failures using 8D, DMAIC, 5 Why, Fishbone and Pareto, closed as verified CAPA.
- Wrote PFMEA, control plans and inspection criteria, plus SOPs for **40+ operations**; trained **60+ operators**.
- Brought in **capital equipment** for robotics and AGV: specs, RFQs, design review, installation, acceptance testing.
- Drove **continuous improvement** with VSM, SMED, 5S, TPM, Kaizen and 4M change control.
- Managed supplier quality across **6 suppliers**, led a **24-technician task force**, tracked OEE and scrap in Power BI.

Delphi TVS Technologies Ltd. — Chennai, India
Graduate Engineer Trainee — Quality Engineering — *Jun 2021 – Sep 2022*

- Checked incoming fuel pump and injector parts to drawing with a **CMM**, precision measurement and **GD&T**.
- Reviewed **non-conforming material**, decided what to do with it, and followed it up with the supplier.

EDUCATION

University of Illinois Chicago — Chicago, IL
Master of Science, Industrial Engineering · *GPA 3.9 / 4.0* — *Aug 2024 – May 2026*
Statistical Quality Control · Operations Research I & II · Work Measurement & Methods · Supply Chain Management

Rajalakshmi Engineering College — Chennai, India
Bachelor of Engineering, Mechanical Engineering · *GPA 3.6 / 4.0* — *Aug 2017 – May 2021*
Total Quality Management · Production Planning & Control · Process Planning & Cost Estimation · CAD/CAM · Metrology

PROJECTS

Field-Failure Root Cause & Error-Proofing · *8D · Cp/Cpk · Poka-Yoke · CAPA* — *Samsung*

- A compressor returned with metal burrs inside; a capability study across **9 stations** traced it to one drilling step.
- Built an in-machine **poka-yoke** air purge so no part leaves the station with burrs; rolled out to similar stations.

Tooling Development — Leak-Test Sealing Cap · *SolidWorks · GD&T · Process Validation* — *Samsung*

- Imported leak-test couplers were damaging parts and causing false rejects at leak test.
- Designed a **SolidWorks** replacement, proved it over a one-month pilot run, and released it into production.

TECHNICAL SKILLS

Process & Equipment Process Development · Process Validation · Equipment Qualification · Commissioning · NPI
Lean & Improvement Lean Manufacturing · Lean Six Sigma Green Belt · Kaizen · 5S · SMED · TPM · Standard Work
Quality & Statistics SPC · Cp/Cpk · Gage R&R · MSA · DOE · PFMEA · Control Plans · CAPA · Non-Conforming Material
Problem Solving Root Cause Analysis · 8D · Fishbone · Pareto · Failure Analysis · Poka-Yoke · Countermeasures
Tools & Systems SolidWorks · AutoCAD · SAP (ERP) · Power BI · Advanced Excel · Minitab · SQL · Python · MS Project